

Bargained Preference Optimization: A Cooperative Solution Concept for Multi-Judge Alignment

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Abstract

Aligning a language model against K conflicting preference judges forces a choice: a single policy is deployed, and it realizes one particular trade-off among the judges. Existing methods commit to this choice implicitly, either averaging the judges or maximizing the worst one. We make the choice itself the object of study. For each judge we define the policy’s *floor*, its worst-case win probability against an adversarial response, and show that the feasible set of floor vectors is a compact convex set with a canonical ideal point at $(\frac{1}{2}, \dots, \frac{1}{2})$. Averaging and maxmin methods select two particular corners of this set, and each violates a basic desideratum: averaging can push a judge below the reference policy’s floor, and maxmin is not covariant to judge recalibration. We instead import the axiomatic bargaining solutions of Nash and of Kalai and Smorodinsky, with the reference policy as the disagreement point, and obtain *Bargained Preference Optimization* (BPO): a concave program whose gradient weights judges by inverse surplus, which trains with the same partition-free log-ratio regression as existing Nash alignment methods, and whose reported floors are valid lower-bound certificates. [TODO: finalize the abstract after the theory and experiment sections exist]

1 Introduction

[TODO: Write after Sections 2–4 stabilize. Planned skeleton: (i) multi-judge alignment ships one policy, so every method is secretly a selection rule on a trade-off surface; (ii) name the surface (the floor set), place averaging and maxmin on it, and show each violates a natural axiom; (iii) the bargaining literature solved exactly this selection problem axiomatically in 1950 and 1975; (iv) contributions: formulation, axiomatic characterization on preference-game domains (Section 5), individual-rationality theorem with the decoy game as the violation certificate for averaging, a practical algorithm with valid certificates, and the panel-cycle study as empirical motivation.]

2 Background

2.1 Preference-based alignment and Nash learning

We work in the preference-based contextual bandit setting that underlies RLHF [Yue et al., 2012, Dudík et al., 2015]. A prompt $x \in \mathcal{X}$ is drawn from a distribution ρ ; a policy maps x to a distribution $\pi(\cdot \mid x) \in \Delta(\mathcal{Y})$ over a finite response set $\mathcal{Y}$; and feedback is relative. A preference oracle $P: \mathcal{Y} \times \mathcal{Y} \times \mathcal{X} \rightarrow [0, 1]$ returns the probability that one response is preferred to another. The oracle is skew-symmetric, $P(y \succ y' \mid x) + P(y' \succ y \mid x) = 1$ and $P(y \succ y \mid x) = \frac{1}{2}$, but need not be induced by any scalar utility: cyclic preferences are allowed [Tversky, 1969, May, 1954]. For policies π, π' we write $P(\pi \succ \pi' \mid x) = \mathbb{E}_{y \sim \pi, y' \sim \pi'}[P(y \succ y' \mid x)]$. All objectives below are

expectations over $x \sim \rho$ of per-context terms, so, following standard practice in Nash learning from human feedback (NLHF) [Munos et al., 2024, Zhang et al., 2025], we state everything for a fixed context and drop x ; every statement lifts by conditioning on x and averaging.

For a single oracle, NLHF seeks a von Neumann winner, a policy no opponent beats in expectation, via the KL-regularized game $\max_{\pi} \min_{\pi'} P(\pi \succ \pi') - \tau \text{KL}(\pi \parallel \mu) + \tau \text{KL}(\pi' \parallel \mu)$ with reference policy μ and temperature $\tau > 0$ [Munos et al., 2024, Azar et al., 2024]. Self-play instantiations such as SPPO and INPO make this practical for language models [Wu et al., 2025, Zhang et al., 2025]; INPO in particular turns the exact mirror-descent step into a partition-free regression on pairwise log-ratios, a device we reuse in Section 4.

Realistic alignment involves K oracles $P_1, \dots, P_K$ (helpfulness, harmlessness, factuality, domain critics) that conflict [Bai et al., 2022, Dai et al., 2024]. Existing treatments commit to a trade-off implicitly: averaging methods collapse the judges into a fixed mixture [Wu et al., 2025, Zhang et al., 2025], maxmin methods harden the worst objective [Chakraborty et al., 2024] or the worst objective weighting at decode time [Son et al., 2026], and per-player formulations give each oracle its own player at the cost of leaving the two-player monotone class [Wu et al., 2026]. This paper asks what these commitments have in common and which one is justified.

2.2 Bargaining problems and solutions

Bargaining theory studies exactly the question above in the abstract: given a set of jointly feasible outcomes and a status quo, which single outcome should K parties agree on? We review the two classical answers from first principles.

Definition 1 (Bargaining problem). A K -player bargaining problem is a pair $(\mathcal{F}, d)$ where $\mathcal{F} \subset \mathbb{R}^K$ is compact and convex, $d \in \mathcal{F}$, the set is comprehensive relative to d (if $v \in \mathcal{F}$ and $d \leq u \leq v$ componentwise then $u \in \mathcal{F}$), and there exists $u \in \mathcal{F}$ with $u > d$ componentwise. A bargaining solution is a map f that assigns to every such problem a point $f(\mathcal{F}, d) \in \mathcal{F}$.

The interpretation: coordinate k of a point $u \in \mathcal{F}$ is the utility player k receives, $\mathcal{F}$ collects every utility vector some joint agreement can realize, and d is the *disagreement point*, what each player gets if negotiation fails. In negotiation language d is the vector of best alternatives to a negotiated agreement. A solution is a rule, defined on all problems at once, and the classical program evaluates rules by axioms rather than case by case.

The Nash solution. Nash [1950] proposed four axioms. *Pareto optimality* (PO): no feasible point dominates the chosen one. *Symmetry* (SYM): if the problem is invariant under permuting players, so is the solution. *Scale covariance* (SC): applying a positive affine map $u_k \mapsto a_k u_k + b_k$ to any player’s utility, with the same map applied to $\mathcal{F}$ and d , transforms the solution by the same map; utilities have no interpersonally comparable units, so the rule must not depend on the units chosen. *Independence of irrelevant alternatives* (IIA): if $\mathcal{F}' \subseteq \mathcal{F}$, d is unchanged, and $f(\mathcal{F}, d) \in \mathcal{F}'$, then $f(\mathcal{F}', d) = f(\mathcal{F}, d)$; deleting options that were not chosen does not change the agreement.

Theorem 2 (Nash, 1950). *Exactly one solution satisfies PO, SYM, SC, and IIA:*

$$f^N(\mathcal{F}, d) = \arg \max_{u \in \mathcal{F}, u \geq d} \prod_{k=1}^K (u_k - d_k) = \arg \max_{u \in \mathcal{F}, u > d} \sum_{k=1}^K \log(u_k - d_k). \quad (1)$$

The maximizer exists and is unique because $\sum_k \log(u_k - d_k)$ is strictly concave on the convex region $\{u \in \mathcal{F} : u > d\}$, which is nonempty by Definition 1. The proof of the characterization is

short and instructive: SC and SYM pin the solution down on the canonical symmetric problem, and IIA transports that answer to an arbitrary problem through the supporting hyperplane of the Nash product at its maximizer; see Nash [1950] or Thomson [1994].

Two readings of (1) matter for us. First, the log form shows the Nash bargaining solution (NBS) is neither the utilitarian point ($\max \sum_k u_k$) nor the egalitarian point ($\max \min_k (u_k - d_k)$): the first-order condition weights player k 's direction by $1/(u_k - d_k)$, the inverse of the realized *surplus* over disagreement, so the player closest to walking away speaks loudest, but no player's gain is traded to exactly zero. Second, because log diverges at zero surplus, the solution is strictly *individually rational* (IR): every player strictly improves on d . Neither corner has this combination, as Section 2.3 makes precise.

The Kalai–Smorodinsky solution. IIA is the contested axiom. It makes the agreement blind to how good each player's opportunities were in the larger set, and Kalai and Smorodinsky [1975] showed this can punish a player whose opportunities improved. Define the *ideal point* of a problem by

$$u_k^*(\mathcal{F}, d) = \max\{u_k : u \in \mathcal{F}, u \geq d\}, \quad (2)$$

the best player k could hope for subject to feasibility and individual rationality.

Example 3 (NBS violates monotonicity). Let $d = (0, 0)$, $\mathcal{F}_A = \text{conv}\{(0, 0), (1, 0), (\frac{3}{4}, \frac{3}{4}), (0, 1)\}$, and $\mathcal{F}_B = \text{conv}\{(0, 0), (1, 0), (1, \frac{7}{10}), (0, 1)\}$. Then $\mathcal{F}_A \subseteq \mathcal{F}_B$ and both problems have ideal point $(1, 1)$, so passing from A to B only adds options and reduces no one's hopes. A direct computation on the two Pareto edges gives $f^N(\mathcal{F}_A, d) = (\frac{3}{4}, \frac{3}{4})$ but $f^N(\mathcal{F}_B, d) = (1, \frac{7}{10})$: player 2's share strictly falls, from 0.75 to 0.70, although the feasible set only grew.

Kalai and Smorodinsky replace IIA by *individual monotonicity* (IM): if $\mathcal{F} \subseteq \mathcal{F}'$, d is unchanged, and $u_j^*(\mathcal{F}, d) = u_j^*(\mathcal{F}', d)$ for every $j \neq i$, then $f_i(\mathcal{F}', d) \geq f_i(\mathcal{F}, d)$. An expansion that leaves the other players' hopes untouched must not hurt player i .

Theorem 4 (Kalai and Smorodinsky, 1975). *For $K = 2$, exactly one solution satisfies PO, SYM, SC, and IM: the maximal feasible point on the segment from d to $u^*(\mathcal{F}, d)$,*

$$f^{\text{KS}}(\mathcal{F}, d) = d + t^*(u^* - d), \quad t^* = \max\{t \geq 0 : d + t(u^* - d) \in \mathcal{F}\}. \quad (3)$$

Lemma 5 (Maximin form). *On comprehensive problems, (3) equals*

$$f^{\text{KS}}(\mathcal{F}, d) \in \arg \max_{u \in \mathcal{F}} \min_k \frac{u_k - d_k}{u_k^* - d_k}, \quad (4)$$

and the optimal value of (4) is t^ .*

Proof. For $u \in \mathcal{F}$ with $u \geq d$ let $t(u) = \min_k (u_k - d_k)/(u_k^* - d_k)$. Then $d + t(u)(u^* - d) \leq u$ componentwise, so comprehensiveness puts it in $\mathcal{F}$, hence $t^* \geq \sup_u t(u)$. Conversely the segment point itself has $t(\cdot) = t^*$. $\square$

So KS equalizes *normalized* surpluses: every player attains the same fraction of the distance from disagreement to their ideal. The normalization $(u_k^* - d_k)$ is not a design choice; it is forced by SC together with IM. In Example 3, KS moves from $(\frac{3}{4}, \frac{3}{4})$ on $\mathcal{F}_A$ to $(\frac{10}{13}, \frac{10}{13})$ on $\mathcal{F}_B$: both players gain from the expansion, as IM demands. For $K \geq 3$ the segment point of (3) remains well defined and weakly Pareto optimal on comprehensive problems, but the four axioms no longer characterize it without further conditions [Roth, 1979]; we adopt (4) as the definition for general K , with a leximin tie-break on degenerate flats, and defer the axiomatic subtleties to Section 5.

Rule	PO	SYM	SC	IIA	IM	strict IR
Utilitarian $\max \sum_k u_k$	✓	✓	×	✓	×	×
Egalitarian $\max \min_k (u_k - d_k)$	weak ^a	✓	×	✓	✓	✓
Nash (NBS)	✓	✓	✓	✓	×	✓
Kalai–Smorodinsky (KS)	✓ ^b	✓	✓	×	✓	✓

Table 1: Axiom satisfaction of the four selection rules on the domain of Definition 1. ^aWeak Pareto optimality; the leximin refinement restores full PO. ^bFor $K \geq 3$, weak PO on comprehensive problems; see the discussion after Lemma 5. Strict IR assumes the nondegeneracy condition of Definition 1.

2.3 Four selection rules and their axioms

The utilitarian rule $\max_{u \in \mathcal{F}} \sum_k u_k$ and the egalitarian rule $\max_{u \in \mathcal{F}} \min_k (u_k - d_k)$ are the two corners the alignment literature already occupies, so it is worth recording how they fare on the same axioms. Both fail SC: rescaling one player’s utility, for instance recalibrating one judge, changes which point they select. The utilitarian rule also fails IR and IM; Example 3 doubles as an IM violation for it, since the utilitarian point moves from $(\frac{3}{4}, \frac{3}{4})$ to $(1, \frac{7}{10})$ as well. The egalitarian rule is the unique rule satisfying strong monotonicity together with weak PO and SYM [Kalai, 1977], which is precisely the axiom bundle that gives up scale covariance. Table 1 summarizes; NBS and KS are the only two rules that keep PO, SYM, and SC, and they part ways only on IIA versus IM.

3 Alignment as a Bargain Among Judges

We now instantiate the abstract objects of Section 2.2 with the alignment problem of Section 2.1. The players are the K preference oracles, which we call judges; the utility of a judge is the worst-case win probability it assigns the deployed policy; and the disagreement point is what each judge gets if alignment fails and the reference model ships unchanged.

3.1 Floors and the feasible floor set

Definition 6 (Floor). For a policy $\pi \in \Delta(\mathcal{Y})$ and judge k , the floor is

$$v_k(\pi) = \min_{\pi' \in \Delta(\mathcal{Y})} P_k(\pi \succ \pi') = \min_{y' \in \mathcal{Y}} P_k(\pi \succ y'), \quad (5)$$

the win probability of π under judge k against an adversarial opponent response; the second equality holds because $P_k(\pi \succ \pi')$ is linear in π' . Write $v(\pi) = (v_1(\pi), \dots, v_K(\pi))$.

Lemma 7 (Basic properties). *Each v_k is concave in π and takes values in $[0, \frac{1}{2}]$. Consequently*

$$\mathcal{F} = \{u \in \mathbb{R}_{\geq 0}^K : u \leq v(\pi) \text{ for some } \pi \in \Delta(\mathcal{Y})\} \quad (6)$$

is compact, convex, and comprehensive, and $d := v(\mu) \in \mathcal{F}$.

Proof. Each v_k is a minimum of linear functions of π , hence concave; $v_k(\pi) \leq P_k(\pi \succ \pi) = \frac{1}{2}$. For convexity of $\mathcal{F}$, if $u \leq v(\pi)$ and $u' \leq v(\pi')$ then concavity gives $\lambda u + (1-\lambda)u' \leq v(\lambda\pi + (1-\lambda)\pi')$. Compactness follows from continuity of v on the compact simplex, and comprehensiveness and $d \in \mathcal{F}$ are immediate from (6). $\square$

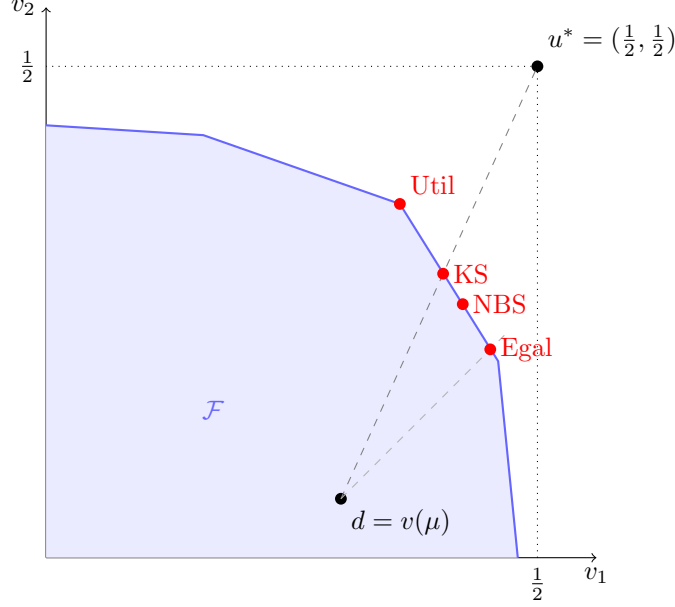


Figure 1: Schematic of the feasible floor set $\mathcal{F}$ for $K = 2$ judges. The ideal point $u^* = (\frac{1}{2}, \frac{1}{2})$ is fixed by Lemma 8 and lies outside $\mathcal{F}$ unless the judges share a von Neumann winner (Corollary 9). Every alignment method deploys one policy and therefore selects one point of $\mathcal{F}$: the utilitarian and egalitarian corners are the existing choices, and the Nash and Kalai–Smorodinsky points are the two selections that survive the axioms of Table 1. The dashed gray line is the KS segment from d to u^* ; the lighter dashed line is the equal-surplus ray defining the egalitarian point.

The floor is the quantity NLHF already cares about: a von Neumann winner of judge k is exactly a maximizer of v_k , and the value it attains is canonical.

Lemma 8 (Canonical ideal point). *For every k , $\max_{\pi} v_k(\pi) = \frac{1}{2}$. More generally, for any weights $w \in \Delta([K])$ the mixture oracle $P_w = \sum_k w_k P_k$ is skew-symmetric, and the scalarized game $\max_{\pi} \min_{\pi'} P_w(\pi \succ \pi')$ has value exactly $\frac{1}{2}$. Hence the ideal point of $(\mathcal{F}, d)$ is $u^* = (\frac{1}{2}, \dots, \frac{1}{2})$ for every d , and u^* dominates all of $\mathcal{F}$.*

Proof. Skew-symmetry of P_w follows term by term since $\sum_k w_k = 1$. A game with skew-symmetric payoff is symmetric, so its value V satisfies $V = 1 - V$ by exchanging the players, giving $V = \frac{1}{2}$; the single-judge claim is the case $w = e_k$, and $v_k \leq \frac{1}{2}$ was shown above. $\square$

Corollary 9. $u^* \in \mathcal{F}$ if and only if the K judges admit a common von Neumann winner, a single policy that is simultaneously maximin for every judge.

Lemma 8 has two consequences worth pausing on. First, the ideal point needs no estimation and no normalization pool: it is fixed by the game structure. Second, scalarization is uninformative as a certificate: whatever the conflict among judges, the averaged game always has value $\frac{1}{2}$, so success against the averaged judge says nothing about any individual floor. Corollary 9 says the only world in which no trade-off exists is the one where all judges already agree on a winner; Condorcet-style impossibility results for preference aggregation indicate this fails generically once panel majorities cycle [Liu et al., 2025]. [TODO: panel-cycle study on real judge panels (Skywork, Athene, ArmoRM pools) as the empirical version of this claim; planned for Section 6]

3.2 Where existing methods sit

Every method that outputs one policy selects one point of $\mathcal{F}$, whether or not it says so. Averaging methods (SPPO-avg, INPO-avg) solve the scalarized game of Lemma 8; in outcome space their idealized counterpart is the utilitarian rule, and the lemma explains why their success metric is vacuous under conflict: the averaged value is $\frac{1}{2}$ no matter how badly a minority judge is served. Maxmin methods, MaxMin-RLHF over reward models [Chakraborty et al., 2024] and adversarial-judge formulations in the small-temperature limit, select the egalitarian corner; RMOD is its decode-time restriction [Son et al., 2026]. Per-player formulations such as heterogeneous MNPO [Wu et al., 2026] do not correspond to a fixed selection rule on $\mathcal{F}$ at all, which is one reading of why their equilibrium analysis is open. Table 1 then reads as a diagnosis: the utilitarian corner can violate individual rationality, serving a judge worse than the untrained reference, and both corners are sensitive to judge recalibration because they fail scale covariance.

We stress that the axioms are engineering desiderata, not claims about negotiating humans. SC is robustness to monotone recalibration of a judge’s scale; IR says no judge falls below the shipped reference; IIA and IM are two forms of stability under changes to the feasible set, for instance under response-pool growth. Which of IIA and IM one prefers determines NBS versus KS, and we develop both.

3.3 Bargained preference optimization

Assumption 10 (Nondegeneracy). There exists π with $v(\pi) > d$ componentwise, i.e., the reference policy is not already Pareto optimal in floor space. (If it fails, we fall back to the leximin refinement of the egalitarian rule; see Remark 13.)

Definition 11 (BPO). The Nash-bargained policy and the Kalai–Smorodinsky-bargained policy are

$$\pi^N \in \arg \max_{\pi \in \Delta(\mathcal{Y})} \sum_{k=1}^K \log(v_k(\pi) - d_k), \quad (7)$$

$$\pi^{KS} \in \arg \max_{\pi \in \Delta(\mathcal{Y})} \min_k \frac{v_k(\pi) - d_k}{\frac{1}{2} - d_k}, \quad (8)$$

with $d = v(\mu)$ and the ideal point $\frac{1}{2}$ supplied by Lemma 8. The regularized variants add $-\tau \text{KL}(\pi \parallel \mu)$ to the objective; (7) then has a unique maximizer by strict concavity.

Both programs are concave: v_k is concave (Lemma 7), $\log$ of a positive concave function is concave, and the KS objective is a minimum of concave functions. There is no saddle point to compute, no inner adversary loop, and no general-sum trap: the conflict among judges is absorbed into the curvature of a single concave objective. By construction π^N realizes the NBS of $(\mathcal{F}, d)$ and π^{KS} realizes the KS point, so Table 1 transfers verbatim to the induced selection rules on floor sets. For $K = 1$ both reduce to maximizing the single floor, recovering the von Neumann winner criterion of NLHF.

Theorem 12 (Individual rationality with teeth). *Under Assumption 10, every judge’s floor strictly exceeds its reference floor at both bargained policies: $v_k(\pi^N) > d_k$ and $v_k(\pi^{KS}) > d_k$ for all k . In contrast, the utilitarian rule admits problems, realizable as K -judge preference games, at which some judge’s floor at the selected policy is strictly below d_k .*

Proof sketch. For (7): Assumption 10 supplies a policy with finite objective value, while the objective tends to $-\infty$ along any sequence whose minimal surplus tends to zero; continuity and compactness give attainment with all surpluses bounded away from zero. For (8): the nondegenerate policy has a strictly positive normalized minimum, so the optimum $t^* > 0$ and every normalized surplus at π^{KS} is at least t^* . The utilitarian violation is by construction: the decoy game of Section 6 exhibits an averaged-oracle optimum whose floor under the hidden judge falls below the uniform reference’s floor. [TODO: promote the decoy instance to a formal example environment and verify the reference-floor comparison numerically] $\square$

Theorem 12 upgrades the familiar hidden-objective collapse from an empirical anecdote to an axiom-violation certificate: averaging does not merely score lower on a metric, it violates individual rationality on a concrete instance, while both bargained policies provably cannot.

Remark 13 (Fallback at degenerate disagreement). If μ is already Pareto optimal in floor space, the NBS is undefined (no positive surplus exists) and KS degenerates. This is the standard boundary case of bargaining theory, and the standard resolution applies: return the leximin-optimal policy, which coincides with μ in floor value when μ is strongly Pareto optimal. One lemma, deferred to Section 5.

Remark 14 (Closest prior work). Nash welfare over *learned scalar reward models* has been analyzed as offline statistical theory [Zhong et al., 2024], and a weighted NBS arises as the limiting characterization of recursive retraining under preference mixtures [Ferbach et al., 2026]. Neither works with preference-game floors, uses the reference policy as the disagreement point, trains an LLM toward the bargained point, or characterizes the selection rule axiomatically on preference-game domains; the Kalai–Smorodinsky solution has, to our knowledge, not appeared in alignment at all. A full comparison is in the related-work section. [TODO: write the related-work section from the July 2026 audit]

4 Practical Algorithm

Three gaps separate Definition 11 from a training loop for language models: the minimum in (5) ranges over all of $\mathcal{Y}$; the floors must be smoothed to admit gradients; and the policy update must avoid partition functions. Each has a standard resolution, and the smoothing we choose has the added property that every reported floor is a valid lower-bound certificate.

4.1 Pool-restricted smoothed floors with valid certificates

As in NLHF practice, the opponent search is restricted to a finite response pool $\mathcal{A}(x) \subseteq \mathcal{Y}$ built from reference, current, and historical policy generations. Write $v_k^{\mathcal{A}}(\pi) = \min_{a \in \mathcal{A}} P_k(\pi \succ a)$; all robustness statements are relative to $\mathcal{A}$ and the K judges. Define the smoothed floor with temperature $\kappa > 0$ by the sum-form log-sum-exp

$$\tilde{v}_k(\pi) = -\kappa \log \sum_{a \in \mathcal{A}} \exp\left(-\frac{P_k(\pi \succ a)}{\kappa}\right). \quad (9)$$

Proposition 15 (Certificate direction). *For every π ,*

$$v_k^{\mathcal{A}}(\pi) - \kappa \log |\mathcal{A}| \leq \tilde{v}_k(\pi) \leq v_k^{\mathcal{A}}(\pi), \quad (10)$$

so $\tilde{v}_k(\pi)$ is a lower bound on the true pool floor, exact as $\kappa \downarrow 0$. Moreover $\tilde{v}_k$ is concave and smooth in π , with gradient

$$\nabla_{\pi} \tilde{v}_k(\pi)(y) = \sum_{a \in \mathcal{A}} q_k(a \mid \pi) P_k(y \succ a), \quad q_k(a \mid \pi) \propto \exp\left(-\frac{P_k(\pi \succ a)}{\kappa}\right). \quad (11)$$

Proof. The sum over $\mathcal{A}$ is at least its largest term $\exp(-v_k^A/\kappa)$ and at most $|\mathcal{A}|$ times it; taking $-\kappa \log$ reverses the inequalities and gives (10). Concavity: each $P_k(\pi \succ a)$ is affine in π , and $c \mapsto -\kappa \log \sum_a e^{-c a/\kappa}$ is concave and coordinatewise nondecreasing, so the composition is concave. The gradient is the chain rule. $\square$

Remark 16 (Why the sum form and not the expectation form). Entropic penalties of KL-distributionally-robust type use the expectation form $-\kappa \log \mathbb{E}_{a \sim s_0} e^{-c a/\kappa}$, which is an *upper* approximation of the minimum: it sits above the hard worst case and certifies nothing about it. The sum form (9) sits below the minimum, so every floor and every surplus this paper reports is a genuine certificate, with slack at most $\kappa \log |\mathcal{A}|$ that the experimenter controls through κ . This distinction is easy to miss and reverses the direction of every downstream guarantee.

The gradient (11) has the right semantics: judge k contributes the win rate of y against a Gibbs opponent q_k concentrated on the pool responses the current policy loses to most under that judge. Each judge fields its own worst-case opponent; there is no single joint adversary and no coupled saddle dynamics.

4.2 The BPO objective and its exact update

Fix the smoothed disagreement point $\tilde{d}_k = \tilde{v}_k(\mu)$, computed once per stage on the same pool, and the surpluses $s_k(\pi) = \tilde{v}_k(\pi) - \tilde{d}_k$. The deployed objectives are

$$J^N(\pi) = \sum_{k=1}^K \log s_k(\pi) - \tau \text{KL}(\pi \parallel \mu), \quad (12)$$

$$J^{\text{KS}}(\pi) = -\beta \log \sum_{k=1}^K \exp\left(-\frac{\sigma_k(\pi)}{\beta}\right) - \tau \text{KL}(\pi \parallel \mu), \quad \sigma_k(\pi) = \frac{s_k(\pi)}{\frac{1}{2} - \tilde{d}_k}, \quad (13)$$

where the outer minimum of (8) is smoothed by the same sum-form device at temperature β , so the reported KS value lower-bounds $\min_k \sigma_k$ within $\beta \log K$. Both objectives are concave (the argument of Proposition 15 applies again), and the τ -KL term makes them strongly concave relative to the entropy, so the maximizer is unique and standard entropic mirror ascent enjoys last-iterate convergence; rates are not this paper's contribution and are deferred. **[TODO: state and prove the convergence proposition; smoothness constants depend on the surplus floor ε below]**

The exact entropic mirror-ascent step on either objective is the familiar multiplicative-weights update with reference mixing,

$$\pi_{t+1}(y) \propto \pi_t(y)^{1-\eta_t\tau} \mu(y)^{\eta_t\tau} \exp(\eta_t g_t(y)), \quad g_t(y) = \sum_{k=1}^K \omega_{t,k} \mathbb{E}_{a \sim q_{t,k}} [P_k(y \succ a)], \quad (14)$$

identical in form to the INPO step, with the single-oracle reward replaced by a surplus-weighted mixture of per-judge Gibbs-opponent win rates. The two solution concepts differ only in the weights:

$$\omega_{t,k}^N \propto \frac{1}{s_k(\pi_t)}, \quad \omega_{t,k}^{\text{KS}} \propto \frac{\exp(-\sigma_k(\pi_t)/\beta)}{\frac{1}{2} - \tilde{d}_k}. \quad (15)$$

Algorithm 1 BPO for prompt-conditioned policies (NBS weights; KS differs only in line 6)

- 1: Initialize $\pi_0 = \mu$. On the stage pool, compute and freeze $\tilde{d}_k = \tilde{v}_k(\mu | x)$ for all k .
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 3: Build or refresh the response pool $\mathcal{A}_t(x)$; score it with the K judges to form $\hat{P}_k$.
 - 4: Estimate pool costs $c_k(a | x) = \hat{P}_k(\pi_t \succ a | x)$ over $\mathcal{A}_t(x)$.
 - 5: Compute $\tilde{v}_k(\pi_t)$ by (9) and surpluses $s_k = \max\{\tilde{v}_k(\pi_t) - \tilde{d}_k, \varepsilon\}$.
 - 6: Set weights $\omega_{t,k} \propto 1/s_k$ (NBS) or $\omega_{t,k} \propto \exp(-\sigma_k/\beta)/(\frac{1}{2} - \tilde{d}_k)$ (KS); normalize.
 - 7: For each judge, sample one opponent $a_k \sim q_{t,k}(\cdot | x) \propto \exp(-c_k(\cdot | x)/\kappa)$.
 - 8: Draw a response pair (y, y') from the pool policy; form the stratified target (17).
 - 9: Update the policy by minimizing $\mathbb{E}[(h_t(\pi; y, y') - \eta_t \hat{g}_t(y, y'))^2]$.
 - 10: **end for**
 - 11: **return** π_T , with certified floors $\tilde{v}_k(\pi_T)$ and surpluses $\tilde{v}_k(\pi_T) - \tilde{d}_k$ per judge.
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NBS weights judges by inverse surplus: a judge close to its reference floor shouts loudest, and the weights are determined by measured quantities rather than by a tunable adversary temperature. KS weights judges by a Gibbs factor on normalized surpluses: the judge furthest behind its ideal-relative pace dominates, with the axiomatically mandated normalization $(\frac{1}{2} - \tilde{d}_k)$ replacing any pool-dependent score normalization. We normalize the weights to sum to one and absorb the overall scale into η_t , which keeps update magnitudes comparable across stages.

4.3 Partition-free regression

Taking log-differences of (14) for two responses cancels the normalizer. With $\Delta_u(y, y') = \log u(y) - \log u(y')$ and

$$h_t(\pi; y, y') = \Delta_\pi(y, y') - \eta_t \tau \Delta_\mu(y, y') - (1 - \eta_t \tau) \Delta_{\pi_t}(y, y'), \quad (16)$$

the exact update satisfies $h_t(\pi_{t+1}; y, y') = \eta_t (g_t(y) - g_t(y'))$, so π_{t+1} is the minimizer of the population regression $\mathbb{E}[(h_t(\pi; y, y') - \eta_t \hat{g}_t(y, y'))^2]$ for any conditionally unbiased estimate $\hat{g}_t$ of the reward gap. We use the judge-stratified estimator: sample one opponent $a_k \sim q_{t,k}$ per judge, independently across judges, and set

$$\hat{g}_t(y, y') = \sum_{k=1}^K \omega_{t,k} [\hat{P}_k(y \succ a_k) - \hat{P}_k(y' \succ a_k)], \quad (17)$$

which is unbiased for $g_t(y) - g_t(y')$ over the independent draws, costs K plug-in evaluations per training pair, gives every judge a signal on every update, and has variance no larger than sampling a single judge-opponent pair from the mixture; the variance decomposition mirrors the stratified-adversary analysis of robust NLHF estimators and is deferred to the appendix. [TODO: port the unbiasedness and variance-reduction propositions with proofs]

At model scale each $\hat{P}_k$ is the usual plug-in from a per-judge reward model through a logistic link, so per-judge transitivity caveats apply exactly as in prior plug-in NLHF work. What disappears is any cross-judge, cross-method score normalization: floors, surpluses, and certificates are all in win-probability units against a fixed pool, the ideal point is the canonical $\frac{1}{2}$, and the evaluation metric is the certified per-judge surplus $\tilde{v}_k(\pi) - \tilde{d}_k$, the same object the algorithm optimizes. The theory and the reported numbers finally coincide.

Implementation notes. (i) *Cost.* Per training pair the target costs K plug-in evaluations, and per stage the pool scoring costs $K|\mathcal{A}|$ evaluations; both match existing stratified adversarial

trainers, so the change relative to an INPO-style codebase is confined to lines 5–7 of Algorithm 1. (ii) *Surplus floor*. The clip at ε in line 5 bounds the NBS weights and the objective’s smoothness constants; if every surplus sits at the clip, Assumption 10 has failed empirically and the leximin fallback of Remark 13 applies. (iii) *Certified IR*. Because $\tilde{d}_k \leq d_k \leq \tilde{d}_k + \kappa \log |\mathcal{A}|$ by (10), a reported surplus above $\kappa \log |\mathcal{A}|$ certifies strict improvement over the true reference floor on the pool; report surpluses together with this margin. (iv) *Ideal point*. We use the canonical $\frac{1}{2}$; a pool-restricted refinement solves one small LP per judge, $\max_{\pi \in \Delta(\mathcal{A})} \min_{a \in \mathcal{A}} \hat{P}_k(\pi \succ a)$, and is worth reporting as a sensitivity check since restricted opponents can only raise the value. (v) *Adversary-free*. There is no inner saddle and no adversary schedule: the Gibbs opponents $q_{t,k}$ and the weights $\omega_{t,k}$ are deterministic functions of the current measured costs, so the analyzed update and the deployed update coincide up to the plug-in and the regression, a mismatch class strictly smaller than in adversarial formulations.

5 Theory: Axiomatic Characterization on Preference-Game Domains

[TODO: The novelty core; to be written next. Planned results: (1) Realizability lemma: every compact convex comprehensive set in the relevant box arises, up to approximation, as the floor set of some K -judge preference game; this makes the domain rich enough for the characterizations to bite. (2) NBS is the unique selection rule on floor sets satisfying PO, SYM, SC of surpluses, and IIA; KS is the unique rule swapping IIA for IM (state precisely for general K , handling the Roth caveat). (3) Leximin fallback lemma for the degenerate case. (4) Certificate concentration: with n Monte Carlo draws per cost entry, Hoeffding plus a union bound over $K|\mathcal{A}|$ entries gives, with probability $1 - \delta$, true floors at least the reported values minus $\kappa \log |\mathcal{A}| + O(\sqrt{\log(K|\mathcal{A}|/\delta)/n})$.]

6 Experiments

[TODO: Planned. (1) Decoy game: reuse the existing synthetic construction; report the four selection rules’ floor vectors and the IR violation of the utilitarian point (certificate for Theorem 12); one figure in the style of Figure 1 with measured points. (2) Panel Condorcet-cycle study: majority relations of the Skywork, Athene, ArmoRM panel over actual generation pools; frequency of cycles and of common winners; motivates Corollary 9 empirically. (3) SafeRLHF conflict: NBS-BPO and KS-BPO versus averaging, maxmin, and offline baselines under the shared-pool protocol; primary metric is the certified per-judge surplus, with the pairwise floor retained for comparability. (4) Ablations: κ and β sweeps against the certificate slack, weight-clipping ε , pool-restricted ideal points.]

7 Related Work

[TODO: Write from the audit memo: NLHF line (Munos, SPPO, INPO, EGPO, Nash-MP, MNPO); robust and multi-objective line (MaxMin-RLHF, group-robust PO, MODPO, RMOD, MOPO, Latham–Moharrami); Blackwell-winner line (Bhatia et al., PROSPER); social-choice-for-alignment line (Ge et al., Liu et al., distortion results); bargaining-adjacent work (Zhong et al., Park et al., Ferbach et al., DISCA, Nash-MTL) with the explicit non-coverage table from the memo.]

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